Nonlinear scalar scattering on Kerr from an endpoint spectral Green-function method

KerrScattering Project[1]

[1] *Manuscript draft generated from the KerrScattering repository*

(Dated: July 6, 2026)

We reformulate the Schwarzschild nonlinear Green-function scattering construction for a weakly self-interacting complex scalar field on Kerr. The logic follows the energy-flux definition of scattering coefficients: solve the linear in problem, use the cubic self-interaction as a source for the first nonlinear field, and read the corrections $T^{(1)}$ and $R^{(1)}$ from the first-order terms in the horizon and infinity fluxes. The scalar Teukolsky equation is solved in the compact coordinate $z = r_+/r$, with infinity and the future horizon retained as collocation endpoints. After the known horizon and far-zone phases are peeled, the degenerate endpoint rows of the radial equation become regular first-order boundary conditions. For $(a, \ell, m, M\omega) = (0.5, 2, 2, 0.30)$ the last 16 normalized Chebyshev coefficients are below 3×10^{-10} on the outer domains and below 2×10^{-13} on the horizon domain; the maximum residual of the original radial equation, normalized by $|\Delta R''| + |\Delta' R'| + |(K^2/\Delta - \lambda)R|$, is 9.2×10^{-10} on the outer domains and 2.4×10^{-11} on the horizon domain. The Kerr factor $\Sigma = r^2 + a^2 \cos^2\theta$ projects the cubic source to $(r^2 C^{(0)}_{\ell'\ell m} + a^2 C^{(2)}_{\ell'\ell m})|R_{\ell m}|^2 R_{\ell m}$, so one incident spheroidal mode sources a target-channel matrix. In the self channel the flux coefficients satisfy $T^{(1)} + R^{(1)} \simeq 0$ after fixed incident normalization, while off-diagonal target channels are first-order field amplitudes whose direct energy-flux contribution begins at quadratic order in the nonlinear correction.

I. INTRODUCTION

Scattering by black holes provides observables complementary to quasinormal mode spectra. In linear theory, scalar, electromagnetic, and gravitational waves on Schwarzschild and Kerr backgrounds exhibit barrier penetration, greybody factors, superradiance, and resonance structures which are well understood through partial waves and related complex-angular-momentum methods [1–7]. These quantities are also relevant for recent frequency-domain descriptions of black-hole ringdowns in terms of greybody factors rather than only sums of quasinormal modes [8, 9].

The nonlinear scattering problem is less developed. The Schwarzschild calculation used as the template for the present work considers a weak self-interaction and defines the nonlinear scattering coefficients in the same way as the linear ones: by the energy flux through the future event horizon and future null infinity. Its central structure is simple. First solve the homogeneous in problem accurately. Next use the linear wave as the source for the first nonlinear field. Finally impose no incoming nonlinear radiation and read the first-order correction from the flux expansion. In this form the nonlinear balance law $T^{(1)} + R^{(1)} = 0$ is a flux statement, not a statement about the raw Green-function amplitudes.

Here we carry this strategy to the scalar Teukolsky equation on Kerr, keeping the same computational philosophy as the Schwarzschild spectral code. The new features are genuinely Kerr-specific. First, the angular basis is spheroidal, not spherical. Second, the radial equation contains the horizon frequency $p = \omega - m\Omega_H$, so the linear flux law allows superradiant reflection. Third, the nonlinear source is not merely a spherical Gaunt coefficient times a radial power: multiplying the equation by $\Sigma = r^2 + a^2 \cos^2\theta$ produces two angular projectors and couples the source mode (ℓ, m) to target channels ℓ' of the same m.

The paper follows the logic of the Schwarzschild nonlinear Green-function template. Section II defines the Kerr scalar problem, the weak nonlinear expansion, and the flux coefficients. Section III describes the endpoint spectral solver and the Green-function construction. Section IV reports spectral accuracy diagnostics, nonlinear self-channel flux balance, and the first target-channel matrix result. The appendices give working derivations used by the code so that normalization and sign conventions can be checked without referring to a separate note.

II. NONLINEAR SCATTERING PROBLEM ON KERR

A. Scalar Teukolsky equation

In Boyer–Lindquist coordinates,

$$\Sigma = r^2 + a^2 \cos^2\theta, \tag{1}$$

$$\Delta = r^2 - 2Mr + a^2 = (r - r_+)(r - r_-), \tag{2}$$

$$r_+ = M + \sqrt{M^2 - a^2}, \qquad r_- = M - \sqrt{M^2 - a^2}, \tag{3}$$

and

$$\Omega_H = \frac{a}{r_+^2 + a^2}. \tag{4}$$

We use the Kerr tortoise coordinate

$$\frac{\mathrm{d}r_*}{\mathrm{d}r} = \frac{r^2 + a^2}{\Delta}. \tag{5}$$

The additive constant in r_* fixes only an overall scattering phase.

For the scalar mode

$$\Phi^{(0)}_{\ell m\omega} = R_{\ell m\omega}(r) S_{\ell m}(\theta; a\omega) e^{\mathrm{i}m\phi - \mathrm{i}\omega t}, \tag{6}$$

the scalar wave operator separates after multiplying by Σ. Substitution gives the identity

$$0 = \frac{1}{R}\frac{\mathrm{d}}{\mathrm{d}r}\left(\Delta\frac{\mathrm{d}R}{\mathrm{d}r}\right) + \frac{K^2}{\Delta} - a^2\omega^2 + 2am\omega$$
$$+ \frac{1}{S\sin\theta}\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\sin\theta\frac{\mathrm{d}S}{\mathrm{d}\theta}\right) + a^2\omega^2\cos^2\theta - \frac{m^2}{\sin^2\theta}. \tag{7}$$

The r-dependent and θ-dependent parts can therefore be set equal to a single separation constant $A_{\ell m}$. With the convention used here, the angular equation is

$$0 = \frac{1}{\sin\theta}\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\sin\theta\frac{\mathrm{d}S_{\ell m}}{\mathrm{d}\theta}\right)$$
$$+ \left[a^2\omega^2\cos^2\theta - \frac{m^2}{\sin^2\theta} + A_{\ell m}\right] S_{\ell m}, \tag{8}$$

with unit normalization on the sphere. The radial equation is

$$\frac{\mathrm{d}}{\mathrm{d}r}\left(\Delta\frac{\mathrm{d}R_{\ell m\omega}}{\mathrm{d}r}\right) + \left(\frac{K^2}{\Delta} - \lambda_{\ell m\omega}\right) R_{\ell m\omega} = 0, \tag{9}$$

where

$$K = (r^2 + a^2)\omega - am, \qquad \lambda_{\ell m\omega} = A_{\ell m} + a^2\omega^2 - 2am\omega. \tag{10}$$

For $a\omega = 0$, $S_{\ell m}$ reduces to the spherical harmonic and $\lambda = \ell(\ell+1)$.

B. Weak nonlinear expansion

We take the self-interacting scalar equation to be

$$\Box\Phi + \epsilon|\Phi|^2\Phi = 0, \qquad |\epsilon| \ll 1, \tag{11}$$

and expand

$$\Phi = \Phi^{(0)} + \epsilon\Phi^{(1)} + \mathcal{O}(\epsilon^2). \tag{12}$$

The zeroth-order field is a single monochromatic spheroidal mode Eq. (9). The first-order field is sourced only by this linear solution; no independent incoming nonlinear radiation is supplied. This is the Kerr analogue of the Schwarzschild setup in which the nonlinear coefficients are extracted from the flux of the perturbative field rather than from a separate scattering experiment.

C. Linear scattering coefficients from energy flux

The horizon-normalized in solution is defined by

$$R^{\mathrm{in}}_{\ell m\omega} \sim e^{-\mathrm{i}pr_*}, \qquad p = \omega - m\Omega_H, \qquad r \to r_+, \tag{13}$$

and at infinity by

$$R^{\mathrm{in}}_{\ell m\omega} \sim B^{\mathrm{inc}}_{\ell m\omega}\frac{e^{-\mathrm{i}\omega r_*}}{r} + B^{\mathrm{ref}}_{\ell m\omega}\frac{e^{+\mathrm{i}\omega r_*}}{r}. \tag{14}$$

With this convention the incident amplitude is B^{inc}, the reflected amplitude is B^{ref}, and the horizon transmission amplitude is unity.

The energy flux is where the scattering coefficients enter. For the complex scalar stress tensor,

$$T_{\mu\nu} = \nabla_{(\mu}\Phi\nabla_{\nu)}\Phi^* - \frac{1}{2}g_{\mu\nu}\nabla_\alpha\Phi\nabla^\alpha\Phi^* + \mathcal{O}(\epsilon), \tag{15}$$

the radial part of the associated mode flux is equivalently encoded in the conserved current

$$J[R] = \frac{\Delta}{2\mathrm{i}}\left(R^*\frac{\mathrm{d}R}{\mathrm{d}r} - R\frac{\mathrm{d}R^*}{\mathrm{d}r}\right). \tag{16}$$

An incoming far-zone wave with amplitude A_{in} carries flux $\omega|A_{\mathrm{in}}|^2$, an outgoing far-zone wave with amplitude A_{out} carries flux $\omega|A_{\mathrm{out}}|^2$, and an ingoing horizon wave with amplitude A_H carries horizon flux $p(r_+^2 + a^2)|A_H|^2$. Therefore the invariant linear reflection and transmission factors for a unit incident wave are

$$R^{(0)} = \left|\frac{B^{\mathrm{ref}}}{B^{\mathrm{inc}}}\right|^2, \qquad T^{(0)} = \frac{p(r_+^2 + a^2)}{\omega|B^{\mathrm{inc}}|^2}, \tag{17}$$

so that

$$T^{(0)} + R^{(0)} = 1. \tag{18}$$

For the horizon-normalized mode, Eq. (16) gives $J_H = -p(r_+^2 + a^2)$ at the future horizon and $J_\infty = -\omega|B^{\mathrm{inc}}|^2 + \omega|B^{\mathrm{ref}}|^2$ at infinity. Hence

$$\omega\left(|B^{\mathrm{inc}}|^2 - |B^{\mathrm{ref}}|^2\right) = p(r_+^2 + a^2), \tag{19}$$

which is Eq. (17). More details are collected in Appendix A. When $p < 0$, $T^{(0)} < 0$ and the reflected flux exceeds the incident flux: this is scalar superradiance.

D. Projected first-order source

For a monochromatic source mode (ℓ, m, ω), the cubic term in Eq. (11) has the same time and azimuthal dependence. We therefore write

$$\Phi^{(1)} = e^{\mathrm{i}m\phi - \mathrm{i}\omega t}\sum_{\ell' \geq |m|} R^{(1)}_{\ell'|\ell m\omega}(r) S_{\ell' m}(\theta; a\omega). \tag{20}$$

Multiplying the scalar wave equation by Σ and projecting onto the target spheroidal harmonic gives

$$\mathcal{L}_{\ell'} R^{(1)}_{\ell'|\ell m\omega} = -Q_{\ell'\ell m}(r), \tag{21}$$

where $\mathcal{L}_{\ell'}$ denotes the radial operator in Eq. (9) with $\lambda_{\ell' m\omega}$, and

$$Q_{\ell'\ell m}(r) = \left[r^2 C^{(0)}_{\ell'\ell m} + a^2 C^{(2)}_{\ell'\ell m}\right]|R^{\mathrm{in}}_{\ell m\omega}|^2 R^{\mathrm{in}}_{\ell m\omega}. \tag{22}$$

The two angular projectors are

$$C^{(s)}_{\ell'\ell m} = \int \mathrm{d}\Omega\, \cos^s\theta\, S^*_{\ell' m}|S_{\ell m}|^2 S_{\ell m}, \qquad s = 0, 2. \quad (23)$$

The factor $r^2 C^{(0)} + a^2 C^{(2)}$ is simply the decomposition of $\Sigma|S_{\ell m}|^2 S_{\ell m}$ into the target basis: the r^2 piece carries the ordinary spheroidal Gaunt coefficient, while the $a^2\cos^2\theta$ piece is a genuinely Kerr correction. This is the first structural difference from the Schwarzschild calculation: even a single incident spheroidal mode sources a target-channel matrix. The self channel $\ell' = \ell$ interferes with the linear field in the boundary fluxes and gives the first nonlinear corrections $T^{(1)}$ and $R^{(1)}$. The off-diagonal channels are genuine first-order fields, but they are orthogonal to the linear outgoing channel in the angular flux and therefore contribute to the total energy flux only at $\mathcal{O}(\epsilon^2)$.

III. METHODS

A. Endpoint pseudo-spectral homogeneous solutions

We solve Eq. (9) on two Chebyshev subdomains in

$$z = \frac{r_+}{r}, \qquad z = 0 \leftrightarrow r = \infty, \qquad z = 1 \leftrightarrow r = r_+. \quad (24)$$

The radial derivatives are converted by

$$\frac{\mathrm{d}}{\mathrm{d}r} = -\frac{z^2}{r_+}\frac{\mathrm{d}}{\mathrm{d}z}, \qquad \frac{\mathrm{d}^2}{\mathrm{d}r^2} = \frac{z^4}{r_+^2}\frac{\mathrm{d}^2}{\mathrm{d}z^2} + \frac{2z^3}{r_+^2}\frac{\mathrm{d}}{\mathrm{d}z}. \quad (25)$$

The known phases are peeled off before discretization,

$$R_{\mathrm{in}} = e^{-\mathrm{i}pr_*}u_{\mathrm{in}}(z), \quad (26)$$

$$R_{\mathrm{down}} = \frac{e^{-\mathrm{i}\omega r_*}}{r}u_{\mathrm{down}}(z), \quad (27)$$

$$R_{\mathrm{up}} = \frac{e^{+\mathrm{i}\omega r_*}}{r}u_{\mathrm{up}}(z). \quad (28)$$

The transformed equation has the schematic form

$$B_2(z)u''(z) + B_1(z)u'(z) + B_0(z)u(z) = 0. \quad (29)$$

At $z = 0$ for down/up modes and at $z = 1$ for the in mode, B_2 vanishes. Thus the endpoint row is a first-order regularity condition rather than an externally imposed truncation. The normalizations are

$$u_{\mathrm{down}}(0) = 1, \qquad u_{\mathrm{up}}(0) = 1, \qquad u_{\mathrm{in}}(1) = 1. \quad (30)$$

For low frequencies we use an analytic sinh mapping to cluster points toward the far-field endpoint, following the same motivation as the Schwarzschild Gevrey-regularity discussion in the base calculation.

At an intermediate point $z_m = r_+/r_m$, the horizon solution is matched to down/up solutions:

$$R_{\mathrm{in}} = B^{\mathrm{inc}}R_{\mathrm{down}} + B^{\mathrm{ref}}R_{\mathrm{up}}. \quad (31)$$

Matching both R and $\mathrm{d}R/\mathrm{d}r$ gives the two-by-two linear system

$$\begin{pmatrix} R_{\mathrm{down}} & R_{\mathrm{up}} \\ R'_{\mathrm{down}} & R'_{\mathrm{up}} \end{pmatrix}_{r_m} \begin{pmatrix} B^{\mathrm{inc}} \\ B^{\mathrm{ref}} \end{pmatrix} = \begin{pmatrix} R_{\mathrm{in}} \\ R'_{\mathrm{in}} \end{pmatrix}_{r_m}. \quad (32)$$

This is the same algebraic matching used in the Schwarzschild spectral route; only the phase factors and the radial operator have changed.

B. Green-function nonlinear amplitudes

Let $R^{\mathrm{in}}_{\ell' m\omega}$ and $R^{\mathrm{up}}_{\ell' m\omega}$ denote the target-channel in and up solutions. The radial Wronskian

$$\mathcal{W}_{\ell'} = \Delta\left(R^{\mathrm{in}}_{\ell'}\frac{\mathrm{d}R^{\mathrm{up}}_{\ell'}}{\mathrm{d}r} - R^{\mathrm{up}}_{\ell'}\frac{\mathrm{d}R^{\mathrm{in}}_{\ell'}}{\mathrm{d}r}\right) \quad (33)$$

is independent of r. The first-order solution is required to contain no incoming nonlinear radiation from infinity and no outgoing nonlinear radiation from the past horizon. The two Green-function amplitudes are therefore

$$A^{(1)}_{\mathrm{ref},\ell'|\ell m} = -\frac{1}{\mathcal{W}_{\ell'}}\int_{r_+}^{\infty} R^{\mathrm{in}}_{\ell'}(r)Q_{\ell'\ell m}(r)\,\mathrm{d}r, \quad (34)$$

$$A^{(1)}_{H,\ell'|\ell m} = -\frac{1}{\mathcal{W}_{\ell'}}\int_{r_+}^{\infty} R^{\mathrm{up}}_{\ell'}(r)Q_{\ell'\ell m}(r)\,\mathrm{d}r. \quad (35)$$

These are the quantities directly computed by the code in unit-horizon normalization.

The sign in Eqs. (34) and (35) comes from $\mathcal{L}R^{(1)} = -Q$. Equivalently, the Green function normalized by the jump condition $\Delta[\partial_r G]^{r=r'-}_{r=r'+} = 1$ is

$$G_{\ell'}(r, r') = \frac{R^{\mathrm{in}}_{\ell'}(r_<)R^{\mathrm{up}}_{\ell'}(r_>)}{\mathcal{W}_{\ell'}}, \quad (36)$$

and $R^{(1)} = -\int GQ\,\mathrm{d}r'$. Taking r to infinity picks out the coefficient of R^{up} and gives Eq. (34); taking r to the future horizon picks out the coefficient of R^{in} and gives Eq. (35).

The outer part of the integrals is highly oscillatory. On the outer domain, R_{in} is a sum of down and up phases, while R_{up} is outgoing. The products in Eqs. (34) and (35) decompose into phase channels $\exp(\mathrm{i}n\omega r_*)$ with $n = -4, -2, 0, 2, 4$. We integrate the exterior tail in r_* with Fourier-weighted quadrature, while the inner finite interval is integrated by Gauss-Legendre quadrature in z.

C. From Green amplitudes to flux coefficients

The amplitudes in Eqs. (34) and (35) correspond to a linear solution with unit horizon transmission. They are not themselves transmission or reflection coefficients. To obtain the physical nonlinear scattering coefficients, we first fix the incident amplitude to unity. The rescaled zeroth-order field is

$$R^{(0)}_{\rm phys} = \alpha R_{\rm in}, \qquad \alpha = \frac{1}{B^{\rm inc}}. \tag{37}$$

Since the source is cubic, the first-order field scales as $|\alpha|^2\alpha$. For the self channel $\ell' = \ell$, the physical reflected and horizon amplitudes are

$$\mathcal{R}_{\rm phys} = \frac{B^{\rm ref}}{B^{\rm inc}} + \epsilon\, \frac{A^{(1)}_{\rm ref}}{|B^{\rm inc}|^2 B^{\rm inc}}, \tag{38}$$

$$\mathcal{H}_{\rm phys} = \frac{1}{B^{\rm inc}} + \epsilon\, \frac{A^{(1)}_{H}}{|B^{\rm inc}|^2 B^{\rm inc}}. \tag{39}$$

Now expand the fluxes defined in Sec. II. At infinity $|\mathcal{R}_{\rm phys}|^2 = R^{(0)} + \epsilon R^{(1)} + \mathcal{O}(\epsilon^2)$, and at the horizon $[p(r_+^2 + a^2)/\omega]|\mathcal{H}_{\rm phys}|^2 = T^{(0)} + \epsilon T^{(1)} + \mathcal{O}(\epsilon^2)$. The first nonlinear flux coefficients are therefore the interference terms

$$R^{(1)} = \frac{2\,{\rm Re}\left[\left(B^{\rm ref}\right)^* A^{(1)}_{\rm ref}\right]}{|B^{\rm inc}|^4}, \tag{40}$$

$$T^{(1)} = \frac{2p(r_+^2 + a^2)}{\omega |B^{\rm inc}|^4}{\rm Re}\, A^{(1)}_H. \tag{41}$$

For a self-channel source, conservation implies

$$T^{(1)} + R^{(1)} = 0, \tag{42}$$

up to numerical error. In the present cubic model this is the first-order part of the conserved $U(1)$ Klein-Gordon flux. Equivalently, the source $Q \propto |R_{\rm in}|^2 R_{\rm in}$ has a real overlap with $R^*_{\rm in}$, so it redistributes the flux between horizon and infinity but does not create a net monochromatic current. The numerical quantity $T^{(1)} + R^{(1)}$ is therefore a direct diagnostic of the combined spectral, matching, and oscillatory-integral error.

IV. RESULTS

A. Spectral accuracy diagnostics

The calculation is assessed with internal spectral diagnostics, following the same philosophy as the Schwarzschild nonlinear Green-function calculation. For the representative absorbing Kerr mode $(a, \ell, m, M\omega) = (0.5, 2, 2, 0.30)$, Table I reports the tail of the Chebyshev expansion and the residual obtained by reconstructing the radial field and substituting it back into the original radial equation at the interior collocation nodes. The residual denominator is the local sum of absolute operator terms, not a coefficient-only scale.

TABLE I. Internal spectral accuracy diagnostics for $(a, \ell, m, M\omega) = (0.5, 2, 2, 0.30)$ at $N = 288$ and $r_m = 40M$. The coefficient tail is the maximum of the last 16 Chebyshev coefficients divided by the largest coefficient on the same subdomain.

branch	coefficient tail	max relative radial residual
$u_{\rm down}$, outer	2.73×10^{-10}	9.16×10^{-10}
$u_{\rm up}$, outer	2.73×10^{-10}	9.16×10^{-10}
$u_{\rm in}$, inner	1.73×10^{-13}	2.30×10^{-11}

Figure 1 shows the same information as a curve rather than as summary statistics. The outer down and up curves lie on top of one another because they differ only by the sign of the peeled far-zone phase. The inner in solution reaches a smaller coefficient floor. The residual normalization in the right panel includes the reconstructed field and its derivatives, so it is invariant under a constant rescaling of the homogeneous solution.

B. Self-channel nonlinear flux coefficients

Table II gives the fixed-incident self-channel coefficients of Eqs. (40) and (41). The Schwarzschild row is kept only as a normalization reference to the base problem. For ordinary absorption ($p > 0$), a positive cubic coupling in the present sign convention increases the transmitted flux and decreases the reflected flux. In the superradiant regime ($p < 0$), $T^{(0)} < 0$ and the signs reverse in the same flux-balance sense.

The nonlinear balance residual $T^{(1)}+R^{(1)}$ is 3.3×10^{-12} for the Schwarzschild row, -5.6×10^{-13} for the absorbing Kerr $a = 0.5$ row, and -2.7×10^{-13} for the high-spin superradiant row. The $a = 0.5$, $M\omega = 0.24$ row is closer to the superradiant threshold and shows a larger absolute residual, -1.1×10^{-8}, because the two nonlinear flux terms are obtained from large nearly cancelling Green-function amplitudes.

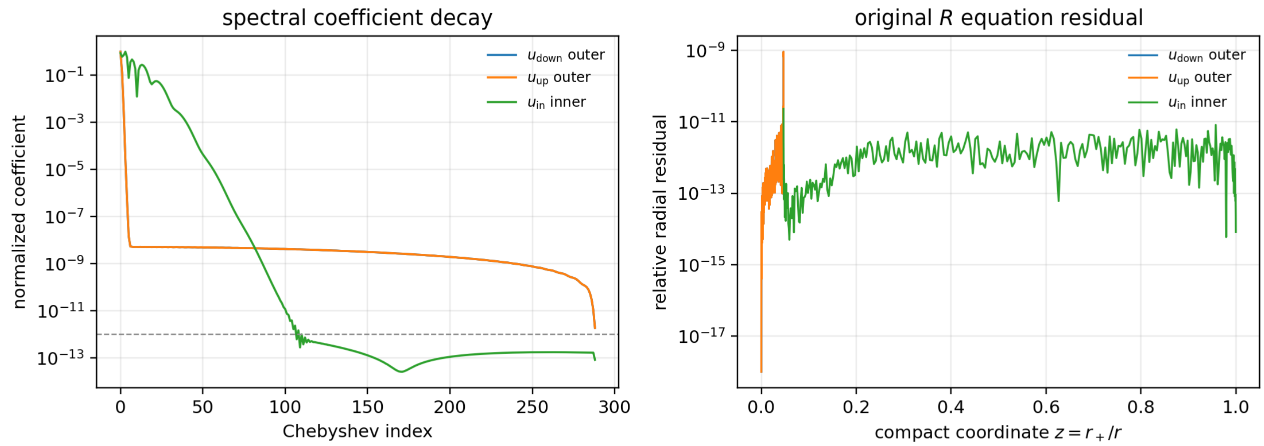

FIG. 1. Spectral quality checks for the representative Kerr mode. Left: normalized Chebyshev coefficient decay after phase peeling. Right: relative residual of the original radial equation at interior collocation nodes, with the denominator $|\Delta R''| + |\Delta' R'| + |(K^2/\Delta - \lambda)R|$.

TABLE II. Self-channel nonlinear flux coefficients in fixed incident normalization. The source model is Eq. (22); computations use $N = 288$, $q = 224$, and Fourier-tail tolerance 10^{-11}.

case	$M\omega$	$T^{(0)}$	$R^{(0)}$	$T^{(1)}$	$R^{(1)}$
Schwarzschild $\ell = 0$	0.10	3.245×10^{-1}	6.755×10^{-1}	8.726×10^{-2}	-8.726×10^{-2}
Kerr $a = 0.5, \ell = m = 2$ superradiant	0.24	-2.373×10^{-6}	1.000002373	-2.153×10^{-7}	2.039×10^{-7}
Kerr $a = 0.5, \ell = m = 2$ absorbing	0.30	1.595×10^{-5}	9.999840×10^{-1}	1.582×10^{-6}	-1.582×10^{-6}
Kerr $a = 0.9, \ell = m = 2$ superradiant	0.58	-4.970×10^{-4}	1.000497031	-7.819×10^{-5}	7.819×10^{-5}

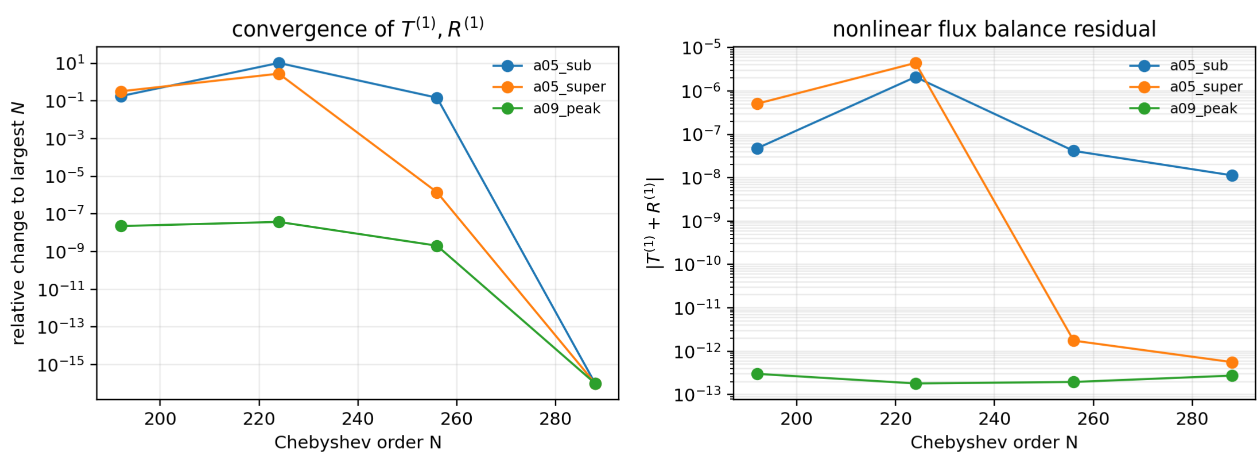

FIG. 2. Self-channel nonlinear flux convergence. Left: relative change of the pair $T^{(1)}, R^{(1)}$ against the largest-N row in the scan. Right: the nonlinear balance residual $|T^{(1)} + R^{(1)}|$. The near-threshold $a = 0.5$, $M\omega = 0.24$ row is a cancellation-limited stress test; the absorbing $a = 0.5$ row and the high-spin superradiant row reach balance residuals at the 10^{-12}–10^{-13} level by $N = 256$–288.

C. Target-channel matrix

For the representative absorbing Kerr mode $(a, \ell, m, M\omega) = (0.5, 2, 2, 0.30)$, the cubic source was projected onto target channels $\ell' = 2, \ldots, 6$. Odd target channels vanish by angular parity in this case. The production computation used $N = 288$, $q = 224$, and Fourier-tail tolerance 10^{-11}.

TABLE III. Unit-horizon first-order Green-function amplitudes for the first Kerr target-channel scan.

ℓ'	$\lvert C^{(0)}_{\ell'\ell m}\rvert$	$\lvert A^{(1)}_{\mathrm{ref},\ell'\lvert\ell m}\rvert$	$\lvert A^{(1)}_{H,\ell'\lvert\ell m}\rvert$
2	1.136×10^{-1}	1.137×10^{6}	3.804×10^{3}
3	0	0	0
4	3.578×10^{-2}	1.362×10^{4}	1.319
5	0	0	0
6	5.341×10^{-3}	2.265×10^{3}	5.383×10^{-7}

A fixed-$q = 224$ convergence scan over $N = 224, 256, 288$ gives a largest reflected-amplitude change of 1.14×10^{-4} relative to the $N = 288$ reference row, while the change from $N = 256$ to $N = 288$ is at most 6.27×10^{-7} over the active reflected channels. The off-diagonal horizon amplitudes are more cancellation limited. The $\ell' = 4$ horizon amplitude is stable at roughly the percent level between $N = 256$ and $N = 288$, whereas $\ell' = 6$ is a near-zero 10^{-7} quantity and should be interpreted only as an order-of-magnitude diagnostic at current double precision.

V. DISCUSSION AND NEXT STEPS

The central conclusion is that the Schwarzschild nonlinear Green-function workflow extends cleanly to Kerr scalar scattering once the spheroidal angular projection and the Kerr flux normalization are handled explicitly. The linear piece is spectrally convergent in the phase-peeled variables and has small original radial-equation residuals on the retained endpoint grid. The nonlinear self-channel reproduces the expected first-order flux balance after converting from unit-horizon to fixed incident normalization. The Kerr geometry also exposes a new object absent from the simple Schwarzschild self-channel calculation: the target-channel amplitude matrix $A^{(1)}_{\mathrm{ref},\ell'|\ell m}$ and $A^{(1)}_{H,\ell'|\ell m}$.

The present results should be read with two normalization caveats. First, the tabulated Green amplitudes are not by themselves physical nonlinear corrections; the physical field correction is $\epsilon A^{(1)}$, and fixed incident normalization introduces the cubic scaling in Eqs. (40)–(41). Second, off-diagonal first-order fields do not interfere with the linear outgoing field in the angular flux, so their energy contribution starts at $\mathcal{O}(\epsilon^2)$.

The immediate numerical extensions are therefore clear: broaden the target-channel scan over (a, ℓ, m, ω), set acceptance gates for the off-diagonal horizon amplitudes, and replace or cross-check the QUADPACK Fourier-tail integral with a dedicated Levin or Filon oscillatory integrator.

Appendix A: Working derivations and normalizations

1. From the scalar wave equation to the radial equation

We use the mostly-plus convention for the Kerr line element. The inverse metric components needed for the scalar wave equation are

$$g^{tt} = -\frac{(r^2+a^2)^2 - a^2\Delta\sin^2\theta}{\Sigma\Delta}, \tag{A1}$$

$$g^{t\phi} = -\frac{2Mar}{\Sigma\Delta}, \tag{A2}$$

$$g^{\phi\phi} = \frac{\Delta - a^2\sin^2\theta}{\Sigma\Delta\sin^2\theta}, \tag{A3}$$

$$g^{rr} = \frac{\Delta}{\Sigma}, \qquad g^{\theta\theta} = \frac{1}{\Sigma}, \qquad \sqrt{-g} = \Sigma\sin\theta. \tag{A4}$$

Thus

$$\Sigma\Box\Phi = \frac{\partial}{\partial r}\left(\Delta\frac{\partial\Phi}{\partial r}\right) + \frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\Phi}{\partial\theta}\right) + \Sigma g^{tt}\frac{\partial^2\Phi}{\partial t^2} + 2\Sigma g^{t\phi}\frac{\partial^2\Phi}{\partial t\,\partial\phi} + \Sigma g^{\phi\phi}\frac{\partial^2\Phi}{\partial\phi^2}. \tag{A5}$$

Substituting $\Phi = R(r)S(\theta)e^{\mathrm{i}m\phi - \mathrm{i}\omega t}$ gives

$$0 = \frac{1}{R}\frac{\mathrm{d}}{\mathrm{d}r}\left(\Delta\frac{\mathrm{d}R}{\mathrm{d}r}\right) + \frac{1}{S\sin\theta}\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\sin\theta\frac{\mathrm{d}S}{\mathrm{d}\theta}\right) + \frac{\omega^2\left[(r^2+a^2)^2 - a^2\Delta\sin^2\theta\right]}{\Delta} - \frac{4Mar\omega m}{\Delta} - \frac{m^2(\Delta - a^2\sin^2\theta)}{\Delta\sin^2\theta}. \tag{A6}$$

Define the frequency-dependent part of Eq. (A6) as

$$\mathcal{T} = \frac{\omega^2\left[(r^2+a^2)^2 - a^2\Delta\sin^2\theta\right]}{\Delta} - \frac{4Mar\omega m}{\Delta} - \frac{m^2(\Delta - a^2\sin^2\theta)}{\Delta\sin^2\theta}. \tag{A7}$$

Using $\Delta = r^2 - 2Mr + a^2$, this can be reorganized as

$$\mathcal{T} = \frac{\left[(r^2+a^2)\omega - am\right]^2}{\Delta} - a^2\omega^2 + 2am\omega + a^2\omega^2\cos^2\theta - \frac{m^2}{\sin^2\theta}. \tag{A8}$$

The first line on the right of Eq. (A8) is radial and the second line is angular. Setting the angular part equal to $-A_{\ell m}$ gives the spheroidal equation in the main text. The radial equation then becomes

$$\frac{\mathrm{d}}{\mathrm{d}r}\left(\Delta\frac{\mathrm{d}R}{\mathrm{d}r}\right) + \left[\frac{K^2}{\Delta} - \left(A_{\ell m} + a^2\omega^2 - 2am\omega\right)\right]R = 0, \tag{A9}$$

which is Eq. (9) with $\lambda = A_{\ell m} + a^2\omega^2 - 2am\omega$.

2. Asymptotic branches and flux balance

Near the future horizon,

$$\Delta = (r_+ - r_-)(r - r_+) + \mathcal{O}\left((r - r_+)^2\right), \tag{A10}$$

$$K(r_+) = (r_+^2 + a^2)p. \tag{A11}$$

Since $\Delta\,\mathrm{d}/\mathrm{d}r = (r^2+a^2)\mathrm{d}/\mathrm{d}r_*$, the leading radial equation is

$$\frac{\mathrm{d}^2R}{\mathrm{d}r_*^2} + p^2R = 0, \qquad r \to r_+. \tag{A12}$$

The two local branches are therefore $e^{\mp\mathrm{i}pr_*}$. The in solution uses the future-horizon ingoing branch $e^{-\mathrm{i}pr_*}$. At infinity the same radial equation reduces to

$$\frac{\mathrm{d}^2(rR)}{\mathrm{d}r_*^2} + \omega^2(rR) = \mathcal{O}(r^{-1})\,rR, \tag{A13}$$

which gives the two branches $e^{\mp\mathrm{i}\omega r_*}/r$.

For any homogeneous solution, the radial part of the Klein-Gordon current is

$$J[R] = \frac{1}{2\mathrm{i}}\Delta\left(R^*\frac{\mathrm{d}R}{\mathrm{d}r} - R\frac{\mathrm{d}R^*}{\mathrm{d}r}\right), \qquad \frac{\mathrm{d}J}{\mathrm{d}r} = 0. \tag{A14}$$

For the horizon-normalized in mode,

$$J_H = -p(r_+^2 + a^2), \tag{A15}$$

$$J_\infty = -\omega|B^{\mathrm{inc}}|^2 + \omega|B^{\mathrm{ref}}|^2. \tag{A16}$$

Equating the two gives Eq. (19). Dividing by the incident energy flux $\omega|B^{\mathrm{inc}}|^2$ yields

$$\frac{p(r_+^2+a^2)}{\omega|B^{\mathrm{inc}}|^2} + \left|\frac{B^{\mathrm{ref}}}{B^{\mathrm{inc}}}\right|^2 = 1. \tag{A17}$$

This also fixes the sign convention for superradiance: if $p < 0$, the horizon energy flux is negative relative to the incident flux and the reflected flux must be larger than the incident flux.

3. Endpoint collocation equations

After the compactification $z = r_+/r$, the horizon is a regular singular endpoint and infinity is an irregular endpoint before phase removal. The phase factors used for the in, down, and up solutions remove precisely the non-analytic oscillatory pieces:

$$R = e^{\mathrm{i}\sigma r_*}r^{-\nu}u(z), \qquad (\sigma,\nu) = (-p,0),(-\omega,1),(+\omega,1), \tag{A18}$$

for in, down, and up solutions respectively. The remaining function has a regular endpoint expansion,

$$u(z) = u_0 + u_1(z - z_e) + u_2(z - z_e)^2 + \cdots, \qquad z_e = 0 \text{ or } 1. \tag{A19}$$

Inserting this expansion into Eq. (29) gives

$$B_1(z_e)u_1 + B_0(z_e)u_0 = 0, \qquad B_2(z_e) = 0. \tag{A20}$$

This is the boundary equation used in the first or last collocation row. One additional row sets the normalization $u_0 = 1$. No artificial horizon cutoff is introduced; the endpoint itself is part of the spectral grid.

4. Green function and nonlinear amplitudes

Let $y_1 = R^{\rm in}_{\ell'}$ and $y_2 = R^{\rm up}_{\ell'}$ solve the target homogeneous equation. For the Sturm-Liouville form

$$\mathcal{L}_{\ell'} y = \frac{\mathrm{d}}{\mathrm{d}r}\left(\Delta \frac{\mathrm{d}y}{\mathrm{d}r}\right) + V_{\ell'}(r) y, \tag{A21}$$

the Wronskian

$$\mathcal{W}_{\ell'} = \Delta(y_1 y_2' - y_2 y_1') \tag{A22}$$

is constant. The Green function

$$G(r, r') = \frac{y_1(r_<) y_2(r_>)}{\mathcal{W}_{\ell'}} \tag{A23}$$

is continuous at $r = r'$ and satisfies

$$\Delta \left[\frac{\partial G}{\partial r}\right]_{r=r'^-}^{r=r'^+} = 1. \tag{A24}$$

Thus $\mathcal{L}_{\ell'} G = \delta(r - r')$. Since the nonlinear radial equation is $\mathcal{L}_{\ell'} R^{(1)}_{\ell'} = -Q_{\ell'}$, the solution satisfying outgoing radiation at infinity and ingoing radiation at the future horizon is

$$R^{(1)}_{\ell'}(r) = -\int_{r_+}^{\infty} G(r, r') Q_{\ell'}(r')\, \mathrm{d}r'. \tag{A25}$$

Taking $r \to \infty$ gives

$$R^{(1)}_{\ell'} \sim A^{(1)}_{\mathrm{ref},\ell'|\ell m} \frac{e^{+\mathrm{i}\omega r_*}}{r}, \tag{A26}$$

$$A^{(1)}_{\mathrm{ref},\ell'|\ell m} = -\frac{1}{\mathcal{W}_{\ell'}} \int_{r_+}^{\infty} y_1 Q_{\ell'}\, \mathrm{d}r, \tag{A27}$$

and taking $r \to r_+$ gives

$$R^{(1)}_{\ell'} \sim A^{(1)}_{H,\ell'|\ell m} e^{-\mathrm{i}p_{\ell'} r_*}, \tag{A28}$$

$$A^{(1)}_{H,\ell'|\ell m} = -\frac{1}{\mathcal{W}_{\ell'}} \int_{r_+}^{\infty} y_2 Q_{\ell'}\, \mathrm{d}r. \tag{A29}$$

These are the formulas used by the production scripts.

5. Why the first-order self-channel balance is expected

For the self channel, the source has the form

$$Q(r) = F(r)|R_{\rm in}|^2 R_{\rm in}, \qquad F(r) = r^2 C^{(0)}_{\ell\ell m} + a^2 C^{(2)}_{\ell\ell m}, \tag{A30}$$

where $F(r)$ is real for the present real-frequency spheroidal basis. Let R_1 solve $\mathcal{L} R_1 = -Q$. Combining this equation with the complex conjugate homogeneous equation for $R^*_{\rm in}$ gives

$$\frac{\mathrm{d}}{\mathrm{d}r}\left[\Delta\left(R^*_{\rm in} R_1' - R_1 R^{*\prime}_{\rm in}\right)\right] = -R^*_{\rm in} Q = -F(r)|R_{\rm in}|^4. \tag{A31}$$

The right-hand side is real. Therefore the imaginary part of the bracket is constant between the horizon and infinity. Evaluating that constant using the asymptotic amplitudes of $R_{\rm in}$ and R_1 gives precisely

$$\frac{2p(r_+^2 + a^2)}{\omega |B^{\rm inc}|^4} \mathrm{Re}\, A^{(1)}_H + \frac{2\,\mathrm{Re}\left[\left(B^{\rm ref}\right)^* A^{(1)}_{\rm ref}\right]}{|B^{\rm inc}|^4} = 0, \tag{A32}$$

which is Eq. (42). The numerical residual in Table II is therefore a direct measure of the combined spectral, matching, and oscillatory-quadrature error.

Appendix B: Schwarzschild limit

For $a = 0$, $r_+ = 2M$, $\Omega_H = 0$, $S_{\ell m}$ reduces to $Y_{\ell m}$, and Eq. (9) is equivalent to the Regge–Wheeler scalar equation for the master variable $\psi = rR$:

$$\left[\frac{\mathrm{d}^2}{\mathrm{d}r_*^2} + \omega^2 - f\left(\frac{\ell(\ell+1)}{r^2} + \frac{2M}{r^3}\right)\right]\psi = 0, \qquad f = 1 - \frac{2M}{r}. \tag{B1}$$

The source Eq. (22) becomes $r^2 C^{(0)}_{\ell'\ell m}|R|^2 R$. In the master variable this is the familiar r^{-2} cubic radial weight of the Schwarzschild nonlinear Green-function problem. Thus the Kerr implementation reduces to the base method when $a = 0$, while retaining the endpoint-collocation boundary treatment.

[1] J. R. Taylor, *Scattering Theory: The Quantum Theory of Nonrelativistic Collisions* (John Wiley and Sons, New York, 1972).

[2] J. A. H. Futterman, F. A. Handler, and R. A. Matzner, *Scattering from Black Holes* (Cambridge University Press, Cambridge, 2012).
[3] S. A. Teukolsky, Perturbations of a rotating black hole. I. Fundamental equations for gravitational, electromagnetic, and neutrino-field perturbations, Astrophys. J. **185**, 635 (1973).
[4] A. A. Starobinsky and S. M. Churilov, Amplification of electromagnetic and gravitational waves scattered by a rotating black hole, Sov. Phys. JETP **38**, 1 (1974).
[5] W. H. Press and S. A. Teukolsky, Floating orbits, superradiant scattering and the black-hole bomb, Nature **238**, 211 (1972).
[6] K. Glampedakis and N. Andersson, Scattering of scalar waves by rotating black holes, Class. Quant. Grav. **18**, 1939 (2001).
[7] S. R. Dolan, Scattering and absorption of gravitational plane waves by rotating black holes, Class. Quant. Grav. **25**, 235002 (2008).
[8] N. Oshita, Thermal ringdown of a Kerr black hole: overtone excitation, Fermi-Dirac statistics and greybody factor, JCAP **04**, 013 (2023).
[9] R. F. Rosato, K. Destounis, and P. Pani, Ringdown stability: Graybody factors as stable gravitational-wave observables, Phys. Rev. D **110**, L121501 (2024).